

# CS 486/686

# Variable Elimination Algorithm

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Lecture 9

RN 13.4 · PM 8.4

# Reminders

✓ **Chrysalis was down earlier today** (server issue) but is **back up now**. It's still offline for the scheduled outage **Thu Jun 11 (9:30 AM) – Fri Jun 12 (noon)**.

📢 **Deadlines still extended: Chat 4 → Tue Jun 16, Chat 5 → Thu Jun 18.**

📝 **Assignment 1 questions → Dake Zhang on Piazza.**

Search ›

Uncertainty ›

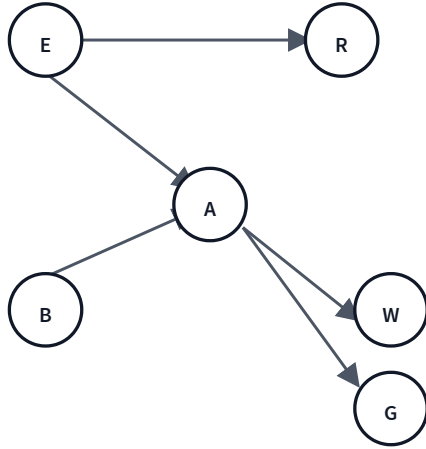
Decisions ›

Learning

## Learning goals

- Do inference in Bayes nets **more efficiently** than the joint
- Define **factors** and the four operations on them
- Trace the **variable elimination algorithm** for prior / posterior queries
- Explain how the **elimination ordering** affects complexity
- Know the basics of **approximate** inference (sampling)

# A query in the Holmes BN: $P(B|w \wedge g)$



*"What's the probability of a burglary, given that both Watson and Gibbon called?"*

- Query variable:  $B$
- Evidence variables:  $W = w, G = g$
- Hidden variables:  $E, A, R$  (to be summed out)

Convention: capital letters are random variables, lowercase letters are specific values.

## From a conditional to a sum

By the definition of conditional probability:

$$P(B \mid w \wedge g) = \frac{P(B, W=w, G=g)}{P(w \wedge g)}$$

The denominator  $P(w \wedge g)$  doesn't depend on  $B$ , so it's just a normalizing constant:

$$P(B \mid w \wedge g) \propto P(B, W=w, G=g)$$

Recover that joint by summing out the hidden variables  $E, A, R$ :

$$P(B, W=w, G=g) = \sum_{e, a, r} P(B, e, a, r, W=w, G=g)$$

# Naive: brute-force sum-out

Apply the BN factorization to that joint:

$$P(B|w \wedge g) \propto \sum_e \sum_a \sum_r P(B) P(e) P(a|B \wedge e) P(w|a) P(g|a) P(r|e)$$

**Q1.** How many additions + multiplications does this require?

- A.  $\leq 10$
- B. 11–20
- C. 21–40
- D. 41–60
- E.  $\geq 61$

**D — 47 ops.** Eight terms to sum, each needing 5 multiplications:  $8 \times 5 + 8 - 1 = 47$  ops.  
Lots of redundant work.

# Push the sums to the right

$P(B)$  doesn't depend on  $e, a, r$  — pull it out.  $P(r|e)$  only depends on  $e, r$  — push  $\sum_r$  inward. And so on:

$$\begin{aligned} P(B) \sum_e P(e) \underbrace{\sum_r P(r|e)}_{=1} \sum_a P(a|B \wedge e) P(w|a) P(g|a) \\ = P(B) \sum_e P(e) \sum_a P(a|B \wedge e) P(w|a) P(g|a) \end{aligned}$$

**Q2.** Ops now?

**B — 14 ops.** Inner sum over  $a$ : 4 mul + 1 add = 5; times 2 for the two values of  $e \Rightarrow 10$ ; multiply each by  $P(e)$ : +2; sum the two  $e$ -terms: +1; multiply by  $P(B)$ : +1. Total = 14.  
~3x speedup just from re-ordering.

# The big idea

- **Reuse** intermediate computations — dynamic programming for probabilities.
- **Exploit** conditional independence baked into the Bayes net.
- Push sums in *as far as possible*, then operate on small tables of probabilities — **factors**.

This is the **Variable Elimination Algorithm**.



# Factors

A **factor**  $f(X_1, \dots, X_j)$  is a function from random-variable assignments to a number. It can represent a joint, a conditional, or a partial assignment.

For the Holmes BN, define one factor per CPT:

$$\begin{aligned} f_1(B) &= P(B), & f_2(E) &= P(E) \\ f_3(A, B, E) &= P(A|B \wedge E), & f_4(R, E) &= P(R|E) \\ f_5(W, A) &= P(W|A), & f_6(G, A) &= P(G|A) \end{aligned}$$

# Restrict: assign a variable

Restricting  $f(X_1, \dots, X_j)$  to  $X_1 = v_1$  fixes that variable, leaving a smaller factor on  $X_2, \dots, X_j$ .

$$\text{Example: } f_1(X, Y) \rightarrow f_2(Y) = f_1(X = t, Y)$$

$f_1(X, Y)$

| X | Y | val |
|---|---|-----|
| t | t | 0.1 |
| t | f | 0.9 |
| f | t | 0.2 |
| f | f | 0.8 |

$X=t$   


$f_2(Y)$

| Y | val |
|---|-----|
| t | 0.1 |
| f | 0.9 |

# Sum out: marginalize a variable

Summing out  $X_1$  from  $f(X_1, \dots, X_j)$  gives a factor on  $X_2, \dots, X_j$  defined by

$$\left(\sum_{X_1} f\right)(X_2, \dots, X_j) = \sum_v f(X_1 = v, X_2, \dots, X_j).$$

Example:  $\sum_Y f_1(X, Y) \rightarrow f_2(X)$

$f_1(X, Y)$

| X | Y | val |
|---|---|-----|
| t | t | 0.1 |
| t | f | 0.4 |
| f | t | 0.3 |
| f | f | 0.5 |

$\xrightarrow{\sum_Y}$

$f_2(X) = \sum_Y f_1$

| X | val             |
|---|-----------------|
| t | 0.1 + 0.4 = 0.5 |
| f | 0.3 + 0.5 = 0.8 |

# Multiply two factors

$(f_1 \times f_2)(X, Y, Z) = f_1(X, Y) \cdot f_2(Y, Z)$  for shared  $Y$  — the product is over the *union* of variables.

| $f_1(X, Y)$ |   |     | $f_2(Y, Z)$ |   |     | $f_1 \times f_2$ |   |   |      |
|-------------|---|-----|-------------|---|-----|------------------|---|---|------|
| X           | Y | val | Y           | Z | val | X                | Y | Z | val  |
| t           | t | 0.1 | t           | t | 0.3 | t                | t | t | 0.03 |
| t           | f | 0.9 | t           | f | 0.7 | t                | t | f | 0.07 |
| f           | t | 0.2 | f           | t | 0.6 | t                | f | t | 0.54 |
| f           | f | 0.8 | f           | f | 0.4 | t                | f | f | 0.36 |
|             |   |     |             |   |     | f                | t | t | 0.06 |
|             |   |     |             |   |     | f                | t | f | 0.14 |
|             |   |     |             |   |     | f                | f | t | 0.48 |
|             |   |     |             |   |     | f                | f | f | 0.32 |

# Normalize a factor

Divide every entry by the sum, so the values sum to 1 (a valid probability distribution).

$f(Y)$

| Y | val |
|---|-----|
| t | 0.2 |
| f | 0.6 |

normalize  
→

$P(Y)$

| Y | val  |
|---|------|
| t | 0.25 |
| f | 0.75 |

$$0.2 / (0.2 + 0.6) = 0.25, 0.6 / 0.8 = 0.75.$$

# The variable elimination algorithm

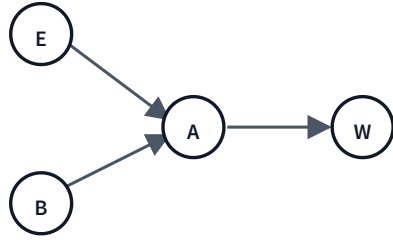
To compute  $P(X_q | X_{o_1} = v_1 \wedge \dots \wedge X_{o_j} = v_j)$ :

## **variable-elimination(BN, query, evidence)**

1. Construct a factor for each CPT in the BN.
2. Restrict every observed variable to its value.
3. For each hidden variable  $X_h$  (in chosen order): multiply all factors that contain  $X_h$ , then sum out  $X_h$  from the product.
4. Multiply the remaining factors.
5. Normalize the result.

Each iteration removes one hidden variable while keeping all factors small.

# Worked example: $P(B|\neg a)$



Sub-BN for the example.  $B$  = query,  $A$  = false = evidence.

**Given CPTs:**

$$P(B) = 0.3, \quad P(E) = 0.1$$

$$P(A|E, B) = 0.8, \quad P(A|E, \neg B) = 0.2$$

$$P(A|\neg E, B) = 0.7, \quad P(A|\neg E, \neg B) = 0.1$$

$$P(W|A) = 0.8, \quad P(W|\neg A) = 0.4$$

**Define factors:**

$$f_1(B), \quad f_2(E), \quad f_3(A, B, E), \quad f_4(W, A)$$

# Eliminate hidden variables

$$f_5(B, E) = f_3(\neg a, B, E)$$

| B | E | val |
|---|---|-----|
| t | t | 0.2 |
| t | f | 0.3 |
| f | t | 0.8 |
| f | f | 0.9 |

$$f_8(B, E) = f_2 \times f_5$$

| B | E | val                     |
|---|---|-------------------------|
| t | t | $0.2 \times 0.1 = 0.02$ |
| t | f | $0.3 \times 0.9 = 0.27$ |
| f | t | $0.8 \times 0.1 = 0.08$ |
| f | f | $0.9 \times 0.9 = 0.81$ |

$$f_9(B) = \sum_E f_8$$

| B | val                  |
|---|----------------------|
| t | $0.02 + 0.27 = 0.29$ |
| f | $0.08 + 0.81 = 0.89$ |



# Multiply remaining factors, normalize

Remaining factors:  $f_1(B)$ ,  $f_9(B)$ ,  $f_7() = 1.0$ . Multiply them:

$$f_{10}(B) = f_1 \times f_9 \times f_7$$

| B | val                                |
|---|------------------------------------|
| t | $0.3 \times 0.29 \times 1 = 0.087$ |
| f | $0.7 \times 0.89 \times 1 = 0.623$ |

normalize

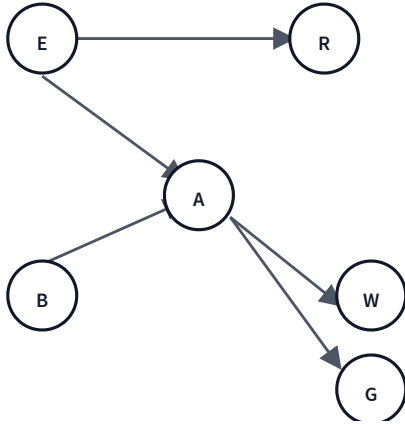


$$P(B|\neg a)$$

| B | val                            |
|---|--------------------------------|
| t | $0.087 / 0.710 \approx 0.1225$ |
| f | $0.623 / 0.710 \approx 0.8775$ |

Burglary is unlikely ( $\approx 12\%$ ) given that the alarm did NOT sound.

# Ordering matters



Compute  $P(G)$  (no evidence). Initial factors:

$$f_1(E), f_2(E, R), f_3(B) \\ f_4(E, B, A), f_5(W, A), f_6(G, A)$$

- **Every** ordering of  $\{R, W, E, B, A\}$  yields a correct algorithm.
- Different orderings produce **different intermediate factors**.
- Larger intermediate factors  $\Rightarrow$  more space + ops.

# Two orderings, very different costs

**Good:**  $R \rightarrow W \rightarrow E \rightarrow$   
 $B \rightarrow A$

- $f_2(E, R) \rightarrow f_7(E)$ : 2 ops
- $f_5(W, A) \rightarrow f_8(A)$ : 2 ops
- $f_1 \cdot f_7 \cdot f_4 \rightarrow f_9(B, A)$ : 20 ops
- $f_9 \cdot f_8 \rightarrow f_{10}(A)$ : 6 ops
- $f_{10} \cdot f_6 \rightarrow f_{11}(G)$ : 6 ops

**Total: 36 ops**

**Bad:**  $A \rightarrow B \rightarrow E \rightarrow R \rightarrow$   
 $W$

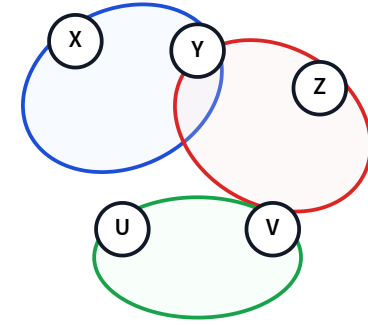
- $f_4 \cdot f_5 \cdot f_6 \rightarrow f_7(E, B, W, G)$ : 80 ops
- $f_7 \cdot f_3 \rightarrow f_8(E, W, G)$ : 24 ops
- $f_1 \cdot f_8 \cdot f_2 \rightarrow f_9(W, G, R)$ : 24 ops
- $f_9 \rightarrow f_{10}(W, G)$ : 4 ops
- $f_{10} \rightarrow f_{11}(G)$ : 2 ops

**Total: 134 ops**

The bad ordering builds a 4-variable intermediate factor  $f_7(E, B, W, G)$  — that's what we pay for.

# Elimination width + tree width

- **Elimination width:** the largest factor (hyperedge) an ordering ever builds.
- **Tree width**  $\omega$  = smallest width over all orderings; VE costs  $2^{O(\omega)}$  time and space.
- Optimal ordering is **NP-hard**, but heuristics do well ( $\omega \approx 8\text{--}10$  even with 1000s of variables).

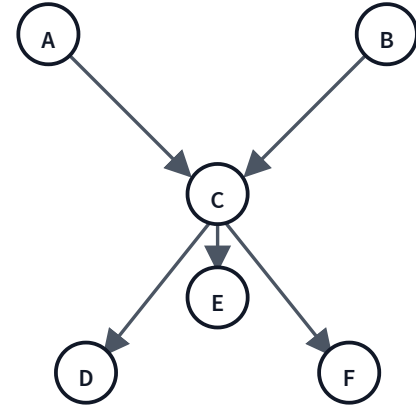


Hyperedges (colored bubbles) = factors.

# Polytrees: an easy case

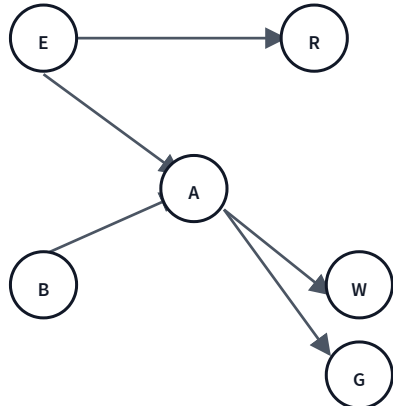
A **polytree** = a **singly-connected** BN: at most one *undirected* path between any two nodes — no loops.

- **Singly-connected** describes the *whole graph*, not one node. Multiple parents are OK:  $C$  below has two ( $A, B$ ) — still a polytree.
- A *diamond* ( $A \rightarrow B \rightarrow D, A \rightarrow C \rightarrow D$ ) is **not** — two routes  $A$  to  $D$ .
- Heuristic: eliminate a **single-neighbor** node (everything but  $C$ ); factors never grow  $\Rightarrow$  VE is **linear**.



$A, B, D, E, F$  each have one neighbor ( $C$ ) — peel those off first;  $C$  (2 parents) last.

# Forward sampling: estimate $P(G)$

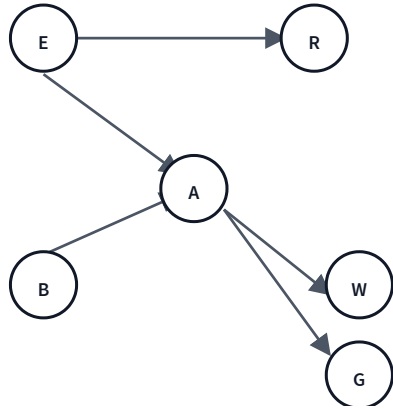


When the BN is too large for exact VE, just **simulate** it.

```
for i in 1..n: ei ~ P(E) bi ~  
P(B) ai ~ P(A | E=ei, B=bi) gi  
~ P(G | A=ai) P(G = T) ≈ (1/n)  
Σi gi
```

Sample in **topological order** so every CPT input is already known. Estimate converges to true  $P(G)$  as  $n \rightarrow \infty$ .

# Rejection sampling: estimate $P(G|\neg b)$



```
for i in 1..n: forward-sample  
  ei, bi, ai, gi if bi == T:  
  reject sample else: keep  $\tilde{g}_i$   
 $P(G = T \mid \neg b) \approx (\sum_i \tilde{g}_i) / \tilde{n}$ 
```

- Only "accepted" samples (consistent with evidence) count.
- **Wasteful** when evidence is rare — most samples thrown away.

## Learning goals (recap)

- ✓ Do inference in Bayes nets **more efficiently** than the joint
- ✓ Define **factors** and the four operations on them
- ✓ Trace the **variable elimination algorithm**
- ✓ Explain how the **elimination ordering** affects complexity
- ✓ Know the basics of **approximate** inference



## Next: Hidden Markov Models

A Bayes net **unrolled over time** — inference about what's happening *now* given a sequence of observations.

VEA in disguise: forward-backward, Viterbi, smoothing.